

RAG Explained

Day 19 — Retrieval-Augmented Generation

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What is RAG?

RAG stands for **Retrieval-Augmented Generation**. It is a technique that allows an AI system to search a company's own documents and information before generating an answer. A normal Large Language Model (LLM) mainly depends on the knowledge it learned during training, so it may not know the latest or private information of a company. RAG solves this problem by connecting the LLM with an external knowledge base.

For example, Petalnex is a software house that provides software/AI services and internship programs. It may have documents containing internship guidelines, project requirements, company policies, service information, FAQs, and technical documentation. Instead of expecting an LLM to remember all this information, a RAG system can search these documents and provide the relevant information to the LLM. The LLM can then generate an answer based on the retrieved information.

RAG Ingestion Process

Before users ask questions, company documents are prepared and stored in a searchable format.

Documents → Extract Text → Chunk Documents → Create Embeddings → Store in Vector Database

Documents can include PDFs, Word files, webpages, FAQs, and company documentation. The text is extracted from these documents and divided into smaller sections called **chunks**. Each chunk is converted into an **embedding**, which represents its meaning as numerical data. These embeddings and related information are then stored in a vector database.

RAG Query Process

When a user asks a question, the system searches the stored knowledge.

User Question → Create Query Embedding → Similarity Search → Retrieve Relevant Chunks → Question + Retrieved Context → LLM → Grounded Answer

For example, an intern might ask, “What are the requirements for the current internship assignment?” The RAG system searches Petalnex's internship documents, retrieves the relevant assignment information, and gives it to the LLM. The LLM then produces an answer based on that retrieved information.

Chunking and Why Chunk Size Matters

Chunking means breaking a large document into smaller pieces before storing it in the vector database. Chunk size is important because chunks that are too large may contain a lot of unrelated information. This can make retrieval less precise and can also send unnecessary information to the LLM.

On the other hand, chunks that are too small may lose important context. For example, if a policy sentence is separated from the conditions that explain it, the AI may not understand the complete meaning.

Therefore, a suitable chunk size should contain enough information to preserve context while remaining focused on one topic. In some RAG systems, chunks may also overlap slightly so that important information at the boundary between chunks is not lost.

3 Real Problems RAG Could Solve at Petalnex

1. Internship Information and Assignment Questions

Interns may repeatedly ask questions about assignment requirements, deadlines, workflow procedures, tools, and submission guidelines. RAG could search the official internship documents and provide consistent answers without requiring staff to answer the same questions repeatedly.

2. Customer Support and Service Information

Petalnex may have information about its software development, AI automation, web development, and other services. A RAG system could retrieve the relevant service documentation and help employees or customers quickly find accurate information about available services, processes, and frequently asked questions.

3. Internal Technical Documentation

Developers and interns may need information about project procedures, APIs, coding guidelines, deployment instructions, or previous project documentation. RAG could search internal technical documents and provide the relevant information, reducing the time employees spend manually searching through files.

Conclusion

RAG is useful for a software house like Petalnex because it connects an LLM with the company's own knowledge. Instead of relying only on general AI knowledge, the system retrieves relevant company information and uses it to generate a grounded answer. This can reduce repetitive questions, improve access to internal knowledge, and help interns, employees, and customers find information more efficiently.